

### C1A: History

C1A Answer the following questions in a few sentences.

- (a) Charles Babbage never succeeded in building his Difference Engine, partly because he had a better idea, for the Analytical Engine. It would have been a programmable computer, the first of its kind in history. Like the Difference Engine, it would have used mechanical gears. To program it, Babbage decided to use a technology used in the textiles industry: what was it, and how was it used in textiles manufacturing?
- (b) On November 4, 1952, the UNIVAC computer was used to predict the result of the Presidential Election between Eisenhower and Stevenson. However, CBS chose not to report its prediction. Why?

**C1B: Hardware**

C1B Answer the following questions in a few sentences.

- (a) Describe a storage device that is designed to be easily connected/disconnected from a computer system **without having to crack open the case or unscrew anything**. (We saw several different examples of them when we took apart the old computers!)

- (b) What is the purpose of the power supply unit (PSU) in a desktop computer?

**C2: Method of Finite Differences**

C2 Consider the function  $f(x) = x^2 + 3$ , for  $x$  taking on whole number values starting at  $x = 0$ .

(a) What is the degree of this function? What information does that give you?

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| $x$ | $f(x)$ | First Diff | Second Diff | Third Diff | Fourth Diff |
|-----|--------|------------|-------------|------------|-------------|
|     |        |            |             |            |             |
|     |        |            |             |            |             |
|     |        |            |             |            |             |
|     |        |            |             |            |             |
|     |        |            |             |            |             |

**C6A: Binary and Hex**

C6A Convert the following:

(a) Convert `0b01010011` to hexadecimal.

(b) Convert `0xF0` to binary.

**C6B: Integer Encodings**

C6B Consider the binary value `0b00001111`.

(a) What value does this represent as a `uint8`?

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C7 Answer the following:

- (a) Suppose NASA sent messages to Mark in binary instead of hexadecimal. They intended to send 0b01010111 but by mistake accidentally sent 0b0111 0111. What's the difference?

| Binary    | Dec | Hex | Char | Binary    | Dec | Hex | Char | Binary    | Dec | Hex | Char | Binary    | Dec | Hex | Char | Binary    | Dec | Hex | Char |
|-----------|-----|-----|------|-----------|-----|-----|------|-----------|-----|-----|------|-----------|-----|-----|------|-----------|-----|-----|------|
| 0010 0000 | 32  | 20  | –    | 0011 0011 | 51  | 33  | 3    | 0100 0110 | 70  | 46  | F    | 0101 1001 | 89  | 59  | Y    | 0110 1100 | 108 | 6C  | l    |
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| 0010 0111 | 39  | 27  | '    | 0011 1010 | 58  | 3A  | :    | 0100 1101 | 77  | 4D  | M    | 0110 0000 | 96  | 60  | `    | 0111 0011 | 115 | 73  | s    |
| 0010 1000 | 40  | 28  | (    | 0011 1011 | 59  | 3B  | ;    | 0100 1110 | 78  | 4E  | N    | 0110 0001 | 97  | 61  | a    | 0111 0100 | 116 | 74  | t    |
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| 0010 1010 | 42  | 2A  | *    | 0011 1101 | 61  | 3D  | =    | 0101 0000 | 80  | 50  | P    | 0110 0011 | 99  | 63  | c    | 0111 0110 | 118 | 76  | v    |
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- (b) As we've discussed, ASCII is limited to 256 characters. What is the extended system called, that allows us to encode everything from Chinese characters to emoji?

**S1: Floating Point**

S1 Encode the number `-2.5` as a `float8`. Be sure to identify the sign bit, exponent bits, and mantissa bits, and put them in the correct order.

CSC-101 Mastery Opportunity 3, 24 September 2026, for **Full Exam**

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C1A Answer the following questions in a few sentences.

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C1B Answer the following questions in a few sentences.

- (a) Describe a storage device that is designed to be easily connected/disconnected from a computer system **without having to crack open the case or unscrew anything**. (We saw several different examples of them when we took apart the old computers!)

- (b) What is the purpose of the power supply unit (PSU) in a desktop computer?

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C2 Consider the function  $f(x) = x^2 + 3$ , for  $x$  taking on whole number values starting at  $x = 0$ .

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|     |        |            |             |            |             |
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|     |        |            |             |            |             |

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|-----------|-----|-----|------|-----------|-----|-----|----------|-----------|-----|-----|----------|-----------|-----|-----|----------|-----------|-----|-----|----------|
| 0010 0000 | 32  | 20  | –    | 0011 0011 | 51  | 33  | <b>3</b> | 0100 0110 | 70  | 46  | <b>F</b> | 0101 1001 | 89  | 59  | <b>Y</b> | 0110 1100 | 108 | 6C  | <b>l</b> |
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| 0010 0010 | 34  | 22  | "    | 0011 0101 | 53  | 35  | <b>5</b> | 0100 1000 | 72  | 48  | <b>H</b> | 0101 1011 | 91  | 5B  | [        | 0110 1110 | 110 | 6E  | <b>n</b> |
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CSC-101 Mastery Opportunity 3, 24 September 2026, for **Devonta Bailey**

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CSC-101 Mastery Opportunity 3, 24 September 2026, for **Xavier Brooks**

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C2 Consider the function  $f(x) = x^2 + 3$ , for  $x$  taking on whole number values starting at  $x = 0$ .

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|-----------|-----|-----|------|-----------|-----|-----|------|-----------|-----|-----|------|-----------|-----|-----|------|-----------|-----|-----|------|
| 0010 0000 | 32  | 20  | –    | 0011 0011 | 51  | 33  | 3    | 0100 0110 | 70  | 46  | F    | 0101 1001 | 89  | 59  | Y    | 0110 1100 | 108 | 6C  | l    |
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| 0010 0000 | 32  | 20  | –    | 0011 0011 | 51  | 33  | <b>3</b> | 0100 0110 | 70  | 46  | <b>F</b> | 0101 1001 | 89  | 59  | <b>Y</b> | 0110 1100 | 108 | 6C  | <b>l</b> |
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- (b) As we've discussed, ASCII is limited to 256 characters. What is the extended system called, that allows us to encode everything from Chinese characters to emoji?

**S1: Floating Point**

S1 Encode the number `-2.5` as a `float8`. Be sure to identify the sign bit, exponent bits, and mantissa bits, and put them in the correct order.

CSC-101 Mastery Opportunity 3, 24 September 2026, for **Arsene Mpanda**

**C6B: Integer Encodings**

C6B Consider the binary value `0b00001111`.

(a) What value does this represent as a `uint8`?

(b) What value does this represent as an `int8`?

**C7: Text Encoding**

C7 Answer the following:

- (a) Suppose NASA sent messages to Mark in binary instead of hexadecimal. They intended to send 0b01010111 but by mistake accidentally sent 0b0111 0111. What's the difference?

| Binary    | Dec | Hex | Char | Binary    | Dec | Hex | Char | Binary    | Dec | Hex | Char | Binary    | Dec | Hex | Char | Binary    | Dec | Hex | Char |
|-----------|-----|-----|------|-----------|-----|-----|------|-----------|-----|-----|------|-----------|-----|-----|------|-----------|-----|-----|------|
| 0010 0000 | 32  | 20  | –    | 0011 0011 | 51  | 33  | 3    | 0100 0110 | 70  | 46  | F    | 0101 1001 | 89  | 59  | Y    | 0110 1100 | 108 | 6C  | l    |
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CSC-101 Mastery Opportunity 3, 24 September 2026, for **Andrew Punzalan**

## C1A: History

C1A Answer the following questions in a few sentences.

- (a) Charles Babbage never succeeded in building his Difference Engine, partly because he had a better idea, for the Analytical Engine. It would have been a programmable computer, the first of its kind in history. Like the Difference Engine, it would have used mechanical gears. To program it, Babbage decided to use a technology used in the textiles industry: what was it, and how was it used in textiles manufacturing?
- (b) On November 4, 1952, the UNIVAC computer was used to predict the result of the Presidential Election between Eisenhower and Stevenson. However, CBS chose not to report its prediction. Why?

**C2: Method of Finite Differences**

C2 Consider the function  $f(x) = x^2 + 3$ , for  $x$  taking on whole number values starting at  $x = 0$ .

(a) What is the degree of this function? What information does that give you?

(b) Compute the setup table for this function. Leave any unneeded entries blank.

| $x$ | $f(x)$ | First Diff | Second Diff | Third Diff | Fourth Diff |
|-----|--------|------------|-------------|------------|-------------|
|     |        |            |             |            |             |
|     |        |            |             |            |             |
|     |        |            |             |            |             |
|     |        |            |             |            |             |
|     |        |            |             |            |             |

**C6A: Binary and Hex**

C6A Convert the following:

(a) Convert **0b01010011** to hexadecimal.

(b) Convert **0xF0** to binary.

**C6B: Integer Encodings**

C6B Consider the binary value `0b00001111`.

(a) What value does this represent as a `uint8`?

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C7 Answer the following:

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| Binary    | Dec | Hex | Char | Binary    | Dec | Hex | Char | Binary    | Dec | Hex | Char | Binary    | Dec | Hex | Char | Binary    | Dec | Hex | Char |
|-----------|-----|-----|------|-----------|-----|-----|------|-----------|-----|-----|------|-----------|-----|-----|------|-----------|-----|-----|------|
| 0010 0000 | 32  | 20  | –    | 0011 0011 | 51  | 33  | 3    | 0100 0110 | 70  | 46  | F    | 0101 1001 | 89  | 59  | Y    | 0110 1100 | 108 | 6C  | l    |
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**C1B: Hardware**

C1B Answer the following questions in a few sentences.

- (a) Describe a storage device that is designed to be easily connected/disconnected from a computer system **without having to crack open the case or unscrew anything**. (We saw several different examples of them when we took apart the old computers!)

- (b) What is the purpose of the power supply unit (PSU) in a desktop computer?

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|-----|--------|------------|-------------|------------|-------------|
|     |        |            |             |            |             |
|     |        |            |             |            |             |
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|     |        |            |             |            |             |

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|-----------|-----|-----|------|-----------|-----|-----|------|-----------|-----|-----|------|-----------|-----|-----|------|-----------|-----|-----|------|
| 0010 0000 | 32  | 20  | –    | 0011 0011 | 51  | 33  | 3    | 0100 0110 | 70  | 46  | F    | 0101 1001 | 89  | 59  | Y    | 0110 1100 | 108 | 6C  | l    |
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| 0011 0000 | 48  | 30  | 0    | 0100 0011 | 67  | 43  | C    | 0101 0110 | 86  | 56  | V    | 0110 1001 | 105 | 69  | i    | 0111 1100 | 124 | 7C  | —    |
| 0011 0001 | 49  | 31  | 1    | 0100 0100 | 68  | 44  | D    | 0101 0111 | 87  | 57  | W    | 0110 1010 | 106 | 6A  | j    | 0111 1101 | 125 | 7D  | }    |
| 0011 0010 | 50  | 32  | 2    | 0100 0101 | 69  | 45  | E    | 0101 1000 | 88  | 58  | X    | 0110 1011 | 107 | 6B  | k    | 0111 1110 | 126 | 7E  | ~    |

- (b) As we've discussed, ASCII is limited to 256 characters. What is the extended system called, that allows us to encode everything from Chinese characters to emoji?

**S1: Floating Point**

S1 Encode the number `-2.5` as a `float8`. Be sure to identify the sign bit, exponent bits, and mantissa bits, and put them in the correct order.

CSC-101 Mastery Opportunity 3, 24 September 2026, for **Dylan Thompson**

## C1A: History

C1A Answer the following questions in a few sentences.

- (a) Charles Babbage never succeeded in building his Difference Engine, partly because he had a better idea, for the Analytical Engine. It would have been a programmable computer, the first of its kind in history. Like the Difference Engine, it would have used mechanical gears. To program it, Babbage decided to use a technology used in the textiles industry: what was it, and how was it used in textiles manufacturing?
- (b) On November 4, 1952, the UNIVAC computer was used to predict the result of the Presidential Election between Eisenhower and Stevenson. However, CBS chose not to report its prediction. Why?

## C2: Method of Finite Differences

C2 Consider the function  $f(x) = x^2 + 3$ , for  $x$  taking on whole number values starting at  $x = 0$ .

- (a) What is the degree of this function? What information does that give you?

- (b) Compute the setup table for this function. Leave any unneeded entries blank.

[illegible]

**C6A: Binary and Hex**

C6A Convert the following:

(a) Convert `0b01010011` to hexadecimal.

(b) Convert `0xF0` to binary.

**C6B: Integer Encodings**

C6B Consider the binary value `0b00001111`.

(a) What value does this represent as a `uint8`?

(b) What value does this represent as an `int8`?

**S1: Floating Point**

S1 Encode the number `-2.5` as a `float8`. Be sure to identify the sign bit, exponent bits, and mantissa bits, and put them in the correct order.

CSC-101 Mastery Opportunity 3, 24 September 2026, for **Jakob Van Pelt**

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**C1B: Hardware**

C1B Answer the following questions in a few sentences.

- (a) Describe a storage device that is designed to be easily connected/disconnected from a computer system **without having to crack open the case or unscrew anything**. (We saw several different examples of them when we took apart the old computers!)

- (b) What is the purpose of the power supply unit (PSU) in a desktop computer?

## C2: Method of Finite Differences

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CSC-101 Mastery Opportunity 3, 24 September 2026, for **Samuel Westfall**

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(a) What is the degree of this function? What information does that give you?

(b) Compute the setup table for this function. Leave any unneeded entries blank.

| $x$ | $f(x)$ | First Diff | Second Diff | Third Diff | Fourth Diff |
|-----|--------|------------|-------------|------------|-------------|
|     |        |            |             |            |             |
|     |        |            |             |            |             |
|     |        |            |             |            |             |
|     |        |            |             |            |             |
|     |        |            |             |            |             |

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